7. **Fire** (6 points)

There is wet wood burning in a fireplace on the ground. Seven meters above ground, the smoke is at a temperature of $t_7 = 40\,^\circ\mathrm{C}$. Disregard the exchange of heat with the surrounding air and assume that the atmospheric pressure at the ground is constant in time and equal to $p_0 = 1000\,\mathrm{hPa}$; the air temperature $t_0 = 20\,^\circ\mathrm{C}$ is independent of height [1]. Assume that the smoke represents an ideal gas of a molar mass $\mu = 29\,\mathrm{g/mol}$ (i.e. equal to the molar mass of the air), and of a molar specific heat at constant volume $C_V = 2.5R$; universal gas constant $R = 8.31\,\mathrm{J/kg \cdot K}$. How high will the smoke column rise?